BONUS Week 3 Homework

This is a preview of the published version of the quiz

Started: Jul 2 at 7:53am

Quiz Instructions

Question 1

1 pts

(Lesson 2.10: Conditional Expectation.) BONUS: Suppose that

$f(x,y) = 6x$ for $0 \leq x \leq y \leq 1$. Hint (you may have seen this someplace): the marginal p.d.f. of X turns out to be

$f_X(x) = 6x(1-x)$ for $0 \leq x \leq 1$. Find the conditional p.d.f. of Y given that $X = x$.

- a. $f(y|x) = \frac{1}{1-x}, \quad 0 \leq x \leq y \leq 1$
- b. $f(y|x) = \frac{1}{1-x}, \quad 0 \leq x \leq 1$
- c. $f(y|x) = \frac{1}{1-y}, \quad 0 \leq y \leq 1$
- d. $f(x|y) = \frac{1}{1-x}, \quad 0 \leq x \leq y \leq 1$

Question 2

1 pts

(Lesson 2.10: Conditional Expectation.) BONUS: Again suppose that

$f(x,y) = 6x$ for $0 \leq x \leq y \leq 1$. Hint (you may already have seen this someplace): the the marginal p.d.f. of X turns out to be $f_X(x) = 6x(1-x)$ for $0 \leq x \leq 1$. Find $\mathbf{E}[Y|X=x]$.

- a. $\mathbf{E}[Y|X=x] = 1/2, \quad 0 \leq x \leq 1$

- b. $\mathsf{E}[Y|X=x] = \frac{1+x}{2}, \quad 0 \le x \le 1$
- c. $\mathsf{E}[Y|X=x] = \frac{1+y}{2}, \quad 0 \le y \le 1$
- d. $\mathsf{E}[X|Y=y] = \frac{1+y}{2}, \quad 0 \le y \le 1$

Question 3

1 pts

(Lesson 2.10: Conditional Expectation.) BONUS: Yet again suppose that $f(x,y) = 6x$ for $0 \le x \le y \le 1$. Hint (you may already have seen this someplace): the the marginal p.d.f. of X turns out to be $f_X(x) = 6x(1-x)$ for $0 \le x \le 1$. Find $\mathsf{E}[\mathsf{E}[Y|X]]$.

- a. 1/2
- b. 2/3
- c. 3/4
- d. 1

Question 4

1 pts

(Lesson 2.14: Estimation.) BONUS: Consider two estimators, T_1 and T_2, for an unknown parameter θ. Suppose that the $\mathsf{Bias}(T_1) = 0$, $Bias(T_2) = \theta$, $\mathsf{Var}(T_1) = 4\theta^2$, and $\mathsf{Var}(T_2) = \theta^2$. Which estimator might you decide to use and why?

- a. T_1 - it has lower expected value
- b. T_1 - it has lower MSE
- c. T_2 - it has lower variance
- d. T_2 - it has lower MSE

Question 5

1 pts

(Lesson 2.15: Maximum Likelihood Estimation.) BONUS: Suppose that $X_1, X_2, \ldots, X_n$ are i.i.d. $\mathrm{Pois}(\lambda)$. Find $\hat{\lambda}$, the MLE of λ. (Don't panic --- it's not that difficult.)

- a. $\overline{X}$
- b. $1/\bar{X}$
- c. $n/\sum_{i=1}^{n} X_i$
- d. S^2

Question 6

1 pts

(Lesson 2.15: Maximum Likelihood Estimation.) BONUS: Suppose that we are looking at i.i.d. Exp(λ) customer service times. We observe times of 2, 4, and 9 minutes. What's the maximum likelihood estimator of λ^2?

- a. 5
- b. 1/5
- c. 25
- d. 1/25

Question 7

1 pts

(Lesson 2.16: Confidence Intervals.) BONUS: Suppose we collect the following observations: $7, -2, 1, 6$ (as in a previous question in this homework). Let's assume that these guys are i.i.d. from a normal distribution with $unknown$ variance σ^2. Give me a two-sided 95% confidence interval for the mean μ.

a. $[-2, 7]$

b. $[-3.75,\ 9.75]$

c. $[-6.75, 6.75]$

d. $[3.75, 9.75]$

Not saved

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